Subset Sums in a Fourfold Sumset

Transported hole bounds and a proof of $N \le 96$

Sum-of-Subset research project

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Abstract

We exclude cardinality 97 and extend the independently audited chain to $71 \le N \le 96$. The proof retains the coarse counting spine of the preceding simplification, transports positive low holes to 14, and merges 412 retained oriented high-hole states into 17 endpoint groups. The reflection in the active-center estimate is fixed by an actual contributor; choosing the numerically stronger orientation is invalid. Only 12 structural leaf groups need nested envelopes. Seventeen integer instances require 35 proved binary disjunctions, each with at least one immediately contradictory child. The resulting 23,303 integer leaf systems are certified by 156 fresh integer vectors: 2,085 symbolic interval checks and 52 tree-leaf checks. Every integer model was also checked separately. The stronger archived bound 84 remains outside the complete independent audit; the present result advances the audited chain rather than surpassing that archived bound.

1 Statement, foundation, and what changes

For a finite real set A, let $\mathrm{FS}(A) = \{\sum_{x\in X} x : X \subseteq A\}$, including zero from the empty subset, and let $4A = A + A + A + A$ allow repeated summands. Let N be the largest possible cardinality under $\mathrm{FS}(A) \subseteq 4A$.

Theorem 1.1. *Every such set has at most 96 elements. Hence* $71 \le N \le 96$.

The lower bound is the unchanged construction $[-31, 31] \cap \mathbb{Z} \cup \{\pm 93, \pm 155, \pm 217, \pm 713\}$. The upper-bound foundation is the independently audited chain $115 \to 103 \to 100 \to 99 \to 98 \to 97$, with the latest endpoint and its simplification in [1, 2]. We reuse its integralization, covering, peeling, inverse, nonprogression, effective-hole, and span lemmas. The archived report [3] provides the strategy of the three disjunctions below. We reconstruct their validity and generate fresh certificates; no archived endpoint certificate is read.

A hypothetical size-97 set contains zero: adjoining zero otherwise gives a forbidden feasible size-98 set. This argument uses the preceding global upper bound; no hereditary feasibility assertion for arbitrary subsets is used. Integralize, remove the four outer elements on each side to obtain D, and peel twice to a core $E = P \cup \{0\} \cup (-Q)$. Write

$$d = |D| = 89, \quad e = |E| = 81, \quad m = a + b = 80, \quad a = |P|,\ b = |Q|, \quad I = (a - 40)^2,$$
$$\Lambda = 1 + T_a + T_b = 1641 + I, \quad \Omega = 22d + 97 - \Lambda = 414 - I, \quad G = 2d - 12 = 166,$$

where $T_r = r(r + 1)/2$. At this endpoint two empty covering units can leave positive low-hole loss 46, larger than the balanced ordinary-quantile threshold 45. Thus the previous ordinary-quantile shortcut must be replaced by core quantiles and, where necessary, proved disjunctions.

2 Lattice and scalar reduction

For a positive r-element set X, retain the definitions

$$\mu(X) = |\mathrm{FS}(X)| - 1 - T_r, \qquad \rho(X) = \mu(X) + 2 - 2\delta_0(X),$$

where $\delta_0(X)$ counts those $x \in X$ for which $2x$ is absent from the restricted two-sum set of $X \cup \{0\}$. Set $\rho = \rho(P) + \rho(Q)$, $\mu = \mu(P) + \mu(Q)$, and $\kappa = |\operatorname{FS}(E)| - \Lambda - \mu$. The inherited nonnegative budget is

$$\rho + 2\theta + c + \mathcal{T} = \Omega, \qquad c = 2|2D| + 18d + 85 - |\operatorname{FS}(D)|, \qquad \mathcal{T} = \kappa + \ell_1 + \ell_2 + g. \tag{1}$$

The one-sided bootstrap is impossible since $1 + T_{81} > 24 \cdot 97 + 77$. The paired bootstrap gives $q^2 \le 2288$, with even q, hence $q \le 46$ and $a, b \ge 17$. Nonnegative Ω leaves 41 oriented pairs, $a = 20, \dots, 60$. For all of them, $T_{\min(a,b)} - 2 - \Omega > 0$, with minimum 131. The simultaneous inverse lemma puts P, Q on a common integer lattice, with side support holes h_P, h_Q satisfying

$$h_P, h_Q \le \kappa, \quad z = h_P + h_Q - \kappa, \quad 0 \le z \le \min(h_P, h_Q) \le \kappa, \quad h_X < |\operatorname{FS}(X)| - 1. \tag{2}$$

The universal span lemma from [1] gives $\max X \le |X| + \rho(X) + h_X + 2$ for $|X| \ge 15$. Thus, with $R_p = \max P$ and $R_q = \max Q$, $R_p + R_q \le m + \rho + \kappa + z + 4$. Before asserting integrality of the peeled layers, the inner comparison is

$$|\operatorname{FS}(D_1)| - 2|\operatorname{FS}(E)| \le C + 2\rho - \mu + \kappa + 2z + \ell_2, \qquad C = 2m + 4e + 14 - \Lambda = -1143 - I.$$

For $z = 0$, its right side is at most $C + 2\Omega = -315 - 3I < 0$. For $z > 0$, both side supports are nonprogressions by (2), so $\rho \ge 52$, $\mu \ge 74$. The right side is at most $C + 3\Omega - \rho - \mu \le -27 - 4I < 0$. An off-lattice point would give two disjoint core-support translates; this contradiction integralizes the inner peeled layer. The outer-layer margin is

$$\Lambda - (4d - 10 + \Omega) = 881 + 2I > 0.$$

The inherited overlap and defect lemmas now apply to the integral set D:

$$w \le 4z, \quad \mathcal{T} \ge \kappa + 166 - 3z, \quad \mathcal{T} \ge \max\{56, z, 166 - 2z\}. \tag{3}$$

Here w counts the support holes of D. If j covering units are empty, put $\delta = c - 89j$. Combining the span lemma with (1)–(3) gives

$$R_p + R_q + \delta \le 332 - I - 89j + 4z. \tag{4}$$

For low holes the scalar budgets are

$$c \le \begin{cases} 248 - I, & z = 0, \\ 196 - I + 2z, & 1 \le z \le 14. \end{cases} \tag{5}$$

Their maxima are 248 and 224, so $j \le 2$. With one empty unit, $\delta \le 159$ or 135, both below $2 \cdot 85$. At most one retained center can therefore have loss at least 85. With two empty units, $\delta \le 70$ or 46, both below 85, so every retained center has more than four hull points. The scalar checker covers all 437 surviving oriented low-hole states.

3 Transport and high-hole grouping

3.1 One monotone change of variables

Let $M_p = \max D$, $M_q = -\min D$, and let S_p, S_q be its positive and negative-magnitude masses. For a retained positive center v, l_v, u_v count points of $v + D$ below 1 and above S_p; reflection defines the negative-side quantities. Always

$$\sum_v (l_v + u_v) \le \delta, \qquad 0 \le l_v + u_v \le 88.$$

Let $K = |2D|$. The cardinality identity gives

$$S_p + S_q + \delta - 2K \le 1686 - 89j + 4z. \tag{6}$$

The mass bonuses are $\varepsilon_p = \sum P - T_a = \mu(P) + h_P$, and similarly for q.

Lemma 3.1 (Endpoint transport)**.** *Suppose a group of states has common lower bounds $\mu(P) \ge u_p$, $\mu(Q) \ge u_q$ and $z \le Z$. Replace*

$$S_p \longmapsto \widehat{S}_p = S_p + Z - z, \qquad S_q \longmapsto \widehat{S}_q = S_q + Z - z.$$

Its necessary linear rows remain valid at the endpoint Z, with mass bonuses $u_p + Z, u_q + Z$, a common upper bound on the group's loss budget, and the span and cardinality right sides evaluated at Z.

Proof. Put $t = Z - z \ge 0$. In a doubled mass row $2M_p + 6R_p - 2S_p + 2z$, or an exposed mass row $\sum_{i=1}^{4} x_i - S_p + z$, the change in the left side is zero. Other mass and upper quantile left sides decrease by t. The left side of (6), after moving $4z$ to the left, decreases by $2t$; the span expression decreases by $4t$. For positive low holes the loss-cap expression $\delta - 2z$ decreases by $2t$. For a high-hole group use the maximum of its scalar caps. All orders, envelope rows, lower quantiles, and losses are unchanged. Nonnegativity is preserved. These auxiliary masses need not arise from another actual set; they provide a necessary relaxation. □

For $1 \le z \le 14$, use $u_p = a - 3$, $u_q = b - 3$ and $Z = 14$. Then $\varepsilon_p \ge a + 11$, $\varepsilon_q \ge b + 11$. The delivered low-hole models have no hole coordinate. With $B_0 = 248 - 89j - I$, their caps and spans become

	δ upper bound	$R_p + R_q + \delta$ upper bound
Z	B_0	$84 + B_0$
P	$B_0 - 24$	$140 + B_0$

The $j = 2$ core condition is selected from the original budget before transport; its validity does not depend on interpreting the transported mass as an actual sum.

3.2 Effective-hole bounds and the fixed orientation

For high holes, use the effective-hole witnesses already proved in [1]. For clarity, the exact witness test is as follows. Given (r, C_*, γ), set

$$K_* = \max\{k \ge 3 : k(k-3)/2 \le C_* + 1\}, \quad D_* = C_* + K_*, \quad E_* = D_* + 1,$$
$$M_* = \max\left\{2\gamma, 1 + \left\lfloor \frac{1 + \sqrt{9 + 8(E_* + 6\gamma)}}{2} \right\rfloor\right\}, \quad J_* = \max\{K_* + 1, 2\gamma + 5, M_*\}.$$

The three strict tests are

$$T_{M_*} - 6\gamma > M_* + 1 + E_*, \quad \gamma(r - J_* + 1) > D_*, \quad T_{J_*-1} - 1 > D_*.$$

A valid witness gives $\mu(X) \ge r + C_* + 1$ when $2\gamma < h_X < |\mathrm{FS}(X)| - 1$. Search $20 \le r \le 64$, $7 \le \gamma \le 20$, $0 \le C_* \le 600$, then for each (r, z) choose the largest valid C_* with $2\gamma < z$, breaking ties by smaller γ. Each used witness is checked again. This finite search makes no optimality claim about the witness parameters.

Write $u_r = r + C_* + 1$. The inherited flexible inverse, for $2 \le \beta \le 10$, is

$$v = \max(0, u_r - 4\beta + 2) = (\beta + 1)q + s, \quad L_\beta(r, u_r) = \min\{(\beta - 1)(r - 14), (\beta - 1)q + \min(s, \beta - 1)\}.$$

Thus set $\rho_r^{\mathrm{low}} = \max\{r - 14, u_r + 2 - 2r, L_2(r, u_r), \ldots, L_{10}(r, u_r)\}$. Every high state satisfies

$$c \le C_a(z) := 414 - I - \rho_a^{\mathrm{low}} - \rho_b^{\mathrm{low}} - \max\{56, z, 166 - 2z\}. \tag{7}$$

The loop $15 \le z \le 414 - I$ is exhaustive because $z \le \mathcal{T} \le \Omega$. It leaves 2,200 nonnegative-budget oriented states, with $C_a(z) \le 118 < 178$. At most one covering unit is empty.

For an all-active configuration, reflect so that the first paired unit's actual contributor is positive, and *then* define a, b. The inherited proof gives one of the two lower bounds

$$F \ge F_A(a,b) + \tfrac{1}{2}(u_a + z) + u_b + z,$$
$$F \ge F_B(a,b) + \tfrac{2}{3}(u_a + z) + u_b + z,$$

where

$$F = S_p + S_q + c - 2(M_p + M_q + R_p + R_q) \le 1728 + 4z,$$
$$F_A = \tfrac{1}{2}T_a + T_b + 6a + 8b + 85, \qquad F_B = \tfrac{2}{3}T_a + T_b + 6a + (22b + 236)/3.$$

Use the minimum of these two differences from $1728 + 4z$ in that fixed orientation. The proof of the active lemma uses nonemptiness to obtain the contributing slots and outer integrality; the hypothesis $c < d$ in its earlier application only served to force this nonemptiness. There is no further need for $c < d$ once all eighteen units are assumed active.

The reflection is not free after a has been assigned. In particular, at $(a, b, z) = (47, 33, 112)$ the valid minimum margin is -1, while the reflected minimum is 377. Taking the larger value would incorrectly remove a necessary model. The checker includes this exact counterexample as a negative control. In the fixed contributor orientation, 117 H0 states remain, with $a = 40, \dots, 47$. There are 295 H1 states with $C_a(z) \ge 89$; for them $\delta \le 29 < \min(a, b) + 5$, so ordinary quantiles are valid.

3.3 Seventeen endpoint groups replace 412 states

Group states with the same (a, j, u_a, u_b), let Z be the largest hole count in a group, and let $C^{\max}$ be the largest $C_a(z)$ there. Lemma 3.1 gives one model with

$$\delta \le C^{\max} - 89j, \quad (\varepsilon_p, \varepsilon_q) \ge (u_a + Z, u_b + Z), \quad R_p + R_q + \delta \le 332 - I - 89j + 4Z.$$

Keep (6) at Z. The complete endpoint table is:

a	u_a	u_b	H0: Z	H0: $C^{\max}$	H1: Z	H1: $C^{\max}$
36	148	224	–	–	64	97
37	156	213	–	–	74	107
38	165	203	–	–	80	113
39	174	193	–	–	84	117
40	184	184	174	4	85	118
41	193	174	173	14	84	117
42	203	165	169	20	80	113
43	213	156	163	23	74	107
44	224	148	153	21	64	97
45	234	139	142	17	–	–
46	245	131	128	10	–	–
47	256	123	112	0	–	–

The scalar enumeration independently reconstructs membership of every group. A further exact row check verifies transport from every one of the 412 retained states to its endpoint. The row types are independent of the outer-center coefficients, so representative modes include both signs and every relevant row type. With four zero-empty modes and seventy one-empty modes, the high model count falls from $117 \cdot 4 + 295 \cdot 70 = 21118$ to $8 \cdot 4 + 9 \cdot 70 = 662$. All 662 endpoint models are excluded by the coarse model below; none needs a nested envelope or binary tree.

4 Models and exhaustive selective refinement

4.1 Covering modes

For the outer magnitudes $p_1 > \cdots > p_4 > M_p$ and their negative reflection $q_1 > \cdots > q_4 > M_q$, define

$$\lambda_p = p_1 - p_2 - p_3 - p_4, \quad \alpha_p = p_2 - p_3 - p_4, \quad \beta_p = 2p_2 - p_1 - p_3 - p_4.$$

The positive center list is

$$v^p = (\lambda_p + p_1, \ldots, \lambda_p + p_4, \lambda_p - q_1, \ldots, \lambda_p - q_4, \alpha_p, \beta_p),$$

with a reflected negative list. String labels `p0`,..., `q9` denote centers, not magnitude variables. Pair 1 is `p4,q4`. In branch A, $\beta_p + \beta_q \le 0$ and pair 2 is `p9,q9`; in branch B, $\beta_p + \beta_q > 0$ and pair 2 is `p5,q5`. The other sixteen covering units are singles.

Enumerate every j-element set of missing units. If pair 1 is present, reflect to choose its positive contributor. If pair 2 is present and nonsplitting, enumerate both contributor choices. In branch A, a missing single `p5` or `q5` can permit pair 2 to split. Drop its individual losses, enumerate an *actually contributing* member $s = p, q$, and retain the necessary row $-\beta_s - M_s \le -1$. If pair 2 itself is missing there is no such member choice. The unchanged pairing argument in [1] proves completeness; there are 4, 70, 578 modes for $j = 0, 1, 2$. The necessary row for a chosen member is never used as a sufficient condition for contribution.

The low root families are:

Family	j	Integer a range	Groups	Integer models
Z0	0	25–55	4	124
P0	0	26–54	4	116
Z1	1	28–52	70	1750
P1	1	29–51	70	1610
Z2	2	32–48	578	9826
P2	2	34–46	578	7514

Together with H0 and H1 this gives 1,966 root groups and 21,602 integer root models. The checker constructs this universe independently of the certificate routing data.

4.2 Coarse and nested rows

All variables are nonnegative. Outer gaps are at least one for $j \le 1$, by the inherited contributing-center integrality argument. For $j = 2$ use only weak outer order. Always $M_s \ge R_s + 4$, $R_p \ge a$, $R_q \ge b$. The branch-B row is $-\beta_p - \beta_q \le -1$ for $j \le 1$, and its weak relaxation ≤ 0 for $j = 2$. The mass rows, on each side, are

$$\begin{aligned} S_p &\ge T_a + \varepsilon_p + 3R_p + 6 + M_p, \\ S_p &\ge T_{a-1} + 4R_p + 6 + M_p. \end{aligned} \tag{8}$$

The first row is multiplied by two in the implementation. Bonuses are zero for Z, $(a + 11, b + 11)$ for P, and $(u_a + Z, u_b + Z)$ for H.

Every pair in $2D$, except pairs of two positive or two negative outliers, lies in $[-M_q - R_q, M_p + R_p]$. There are at most ten unordered repeated outlier pairs of each sign. Thus

$$K \le M_p + M_q + R_p + R_q + 21. \tag{9}$$

Twice this row added to (6) eliminates K:

$$S_p + S_q + \delta - 2(M_p + M_q + R_p + R_q) \le 1728 - 89j + 4Z, \tag{10}$$

where $Z = 0$ in Z, $Z = 14$ in P, and Z is the high endpoint in H. The loss sum, caps, joint span, orders, mass rows and quantiles complete the coarse model. Unlike the previous simplification, the loss cap and span rows are retained: some certificates at this endpoint use them.

For a positive center the necessary endpoint rows are:

Type	Lower	Upper
Global	$-v - l_v - M_p \le -89$	$v - u_v - S_p - M_q \le -88$
Core	$-v - l_v - R_p \le -85$	$v - u_v - S_p - R_q \le -84$
Ordinary	$-v - l_v \le -b - 5$	$v - u_v - S_p \le -a - 4$

Global rows hold for every nonempty center by integer interval counting. Core rows hold if its hull count exceeds four, since only four elements on the relevant side are outside the core; use the core-gap count from [1]. Ordinary rows apply at the relevant in-hull endpoint. Use the reflected rows for negative centers. Individual slot caps are $\lambda_v \le 84$ in a core case, and $\lambda_v \le 88$ otherwise.

Initially use global quantiles, except for H1 where ordinary rows have already been justified. If a two-empty root needs refinement, use core quantiles throughout, including P2. A one-empty low root can be refined exhaustively into: no loss ≥ 85, or each retained center in turn is the unique large-loss center. In the former case use core rows everywhere; in the latter impose $\lambda_v \ge 85$, use global rows there, and core rows elsewhere. The large-center ranges are ZL: $32 \le a \le 48$ and PL: $33 \le a \le 47$, from (5) and $\delta \ge 85$.

Only six roots need that split. They are Z1 branch A with pair 1 missing and either pair-2 contributor chosen, Z1/P1 branch A with pair 2 missing, and Z1/P1 branch A with single `p0` missing and the negative pair-2 contributor chosen. Each has 17 retained centers and 18 outcomes. All other one-empty roots are excluded without naming a large-loss center.

Only twelve resulting structural groups need the nested model. Expose $x_0 = R_p < x_1 < x_2 < x_3 < x_4 = M_p$ and its negative analogue y_i, with integral gaps at least one, and add the mass rows

$$S_p \ge T_a + \varepsilon_p + \sum_{i=1}^{4} x_i, \qquad S_p \ge T_{a-1} + \sum_{i=0}^{4} x_i,$$

with their reflections. For $t = 1, 2, 3$, set $W_t = x_t + y_t$ and $s_t = 2(4 - t)$. The inherited interval-envelope proof gives

$$K \le 2W_t + 1 + \binom{s_t + 1}{2} + \sum_{i=t+1}^{4} \left[\min(W_t + 1, x_i - x_{i-1}) + \min(W_t + 1, y_i - y_{i-1})\right].$$

Indeed the inner hull's double sum has $2W_t + 1$ positions, successive external translates add at most their gap or the hull length, and pairs among the remaining external points account for the binomial term. Linearize each minimum by a nonnegative variable bounded above by both arguments; an actual set chooses the minimum itself. Keep K, (9), and (6) in this model. Levels 0 and 4 are unnecessary. Two extra nonnegative loss coordinates are reserved for the sole-member disjunction below.

5 Three disjunctions, including transported upper endpoints

The disjunctions are necessary for actual configurations. They need not partition every point of a relaxed real polyhedron. Each child is checked; no child is discarded on the basis of a numerical solver status. Every retained contributing center is integral: a contributing point belongs both to a translate of integral D and to the integral target support. The original masses, z, and the chosen endpoints Z are integers, so the transported masses are integers as well. These facts justify the unit gaps below for actual configurations, even though the linear relaxations allow real coordinates.

Lemma 5.1 (Ordinary endpoint or outside endpoint)**.** *For a retained positive center v, with S_p either the actual or a transported larger mass, the following disjunctions hold:*

$$\begin{aligned} -v - l_v \le -b - 5 \quad &or \quad (l_v \ge b + 5 \text{ and } v \le -1), \\ v - u_v - S_p \le -a - 4 \quad &or \quad (u_v \ge a + 5 \text{ and } v \ge S_p + 1). \end{aligned}$$

Their reflections hold on the negative side.

Proof. If $v \ge 0$, at most v of the $b + 5$ nonpositive elements of D can remain above the lower endpoint, giving the first ordinary row. Otherwise the integral center satisfies $v \le -1$ and all these elements are lost. At the upper end, when $v \le S_p$, at most $S_p - v$ positive integers can fit below

the endpoint, so the upper ordinary row holds. When $v > S_p$, integrality gives $v \geq S_p + 1$; zero and all $a + 4$ positive points are lost. For transported masses, interpret the count in the enlarged hull. The recorded upper loss from the original hull is at least the new upper loss, so the ordinary bound remains valid. If $v > S_p$ for the enlarged hull, it is also beyond the original endpoint, so at least $a + 5$ original points are lost. One must select the alternative using the enlarged endpoint, not carry an old outside assertion through the transport without checking it. □

Lemma 5.2 (Inner extreme or endpoint loss)**.** *For $i = 1, 2, 3, 4$, the same center satisfies*

$$l_v \geq i \quad or \quad v \geq y_{5-i} + 1, \qquad u_v \geq i \quad or \quad v \leq S_p - x_{5-i}.$$

Proof. If fewer than i points are lost at the lower end, at least one of the i largest negative magnitudes lies above it. The smallest of those i is y_{5-i}, yielding $v - y_{5-i} \geq 1$. If fewer than i are lost above the original upper endpoint, at least one of the i largest positive points lies below it, giving $v + x_{5-i} \leq S_p^{\text{original}} \leq S_p$. The loss counts are integers, so these two alternatives are exhaustive. □

Lemma 5.3 (Both members or the selected sole member)**.** *In a two-empty mode with a potentially split pair 2, let s be the member chosen from the actual contributors, and let t be the other side. Either $\beta_t + M_t \geq 1$, or there exist hull losses $l_*, u_* \geq 0$ for the selected member satisfying*

$$\sum_{v \text{ retained}} (l_v + u_v) + l_* + u_* \leq \delta,$$
$$-\beta_s - l_* - R_s \leq -85, \qquad \beta_s - u_* - S_s - R_t \leq -84.$$

Proof. If the other member contributes, the first necessary inequality holds. Otherwise the selected member is the sole contributor. Its unit's capacity loss is at least its individual hull loss, so the strengthened loss sum holds. That member has at least $89 - \delta > 4$ hull points, since the two-empty budgets are at most 70 and 46. Its core quantiles follow. Enlarging the upper mass only weakens its upper row. The selection of s from actual contributors, rather than from a merely necessary inequality, is essential to the sole-member conclusion. □

All seventeen integer instances that use a tree appear below. A range in a cell means each integer separately; the displayed node counts correspond to those integers in order. All are in branch A. A large center is present only in ZL and PL and is always `p8`.

Family	Missing units	Choice or actual member	a	Internal nodes
ZL	`p0`	negative contributor	38,39	1,9
PL	`p0`	negative contributor	37–40	1,1,1,1
Z2	`p0,p8`	negative contributor	38,39,40	1,10,2
Z2	`p0,q5`	negative actual member	40–42	1,1,1
P2	`p0,p8`	negative contributor	37–40	1,1,1,1
P2	`p0,q5`	negative actual member	40	1

There are 35 internal nodes: 22 inner-extreme, nine ordinary/outside, and four both/sole-member splits. Every internal node has at least one terminal child; the longest remaining chain has ten splits. The 17 trees have $17 + 35 = 52$ leaves. This structure is discovered numerically, but its exhaustiveness and every leaf contradiction are checked exactly.

6 Exact certificates, validation, and conclusion

For each structural group, all model coefficients are integers and the right side is an exact rational quadratic polynomial in a. Every stored row-weight vector u consists of positive integers. For a model $Ax \leq b(a)$, $x \geq 0$, the checker requires

$$uA \geq 0 \quad \text{coordinatewise}, \qquad \max_{a \in [L,U] \cap \mathbb{Z}} ub(a) < 0.$$

The integer maximum uses endpoints and, for a concave quadratic, the floor and ceiling of its vertex when in range. Polynomials are built symbolically rather than inferred from samples. Trees occur only at explicit integer parameters and use exact integer row combinations.

Proof component	Count
Root groups / integer root models	1966 / 21602
Direct coarse root groups	1900
Two-empty coarse / nested root refinements	51 / 9
One-empty roots requiring a loss split	6
Coarse / nested outcomes of those loss splits	105 / 3
Final structural groups / nested groups	2068 / 12
Integer parameter models before binary trees	23268
Tree roots / internal nodes / tree leaves	17 / 35 / 52
Integer leaf systems after all refinements	23303
Symbolic interval checks / tree-leaf checks	2085 / 52
Fresh integer weight vectors	156
Least normalized strict gap	2/299

The last integer-system count replaces each branched parameter by all its leaves, hence $23268+35 = 23303$. It should not be confused with the number of distinct matrix combinations: interval checking avoids their pointwise repetition. The separate high-state grouping replaces 21,118 potential high models by 662 without omitting an actual state.

The verifier independently constructs all roots, loss outcomes, and allowed disjunctions. It rejects a missing root or member choice, omitted large-loss positions, gaps and overlaps between intervals and tree parameters, missing children, repeated path splits, unknown disjunctions, invalid row weights, and unjustified model types. It checks high endpoint transport on all retained states. Verification uses only the Python standard library and explicit checks, which remain active under `-O`. The optional SciPy generator merely discovers candidates; only exact integer checks permit a certificate to be saved.

The delivered `-exhaustive` run rebuilds all 23,268 integer parameter models, compares their rows with the symbolic formulas, verifies the assigned certificates and all tree leaves, and checks that each interval's worst gap matches the pointwise minimum. Its output agrees with the interval run except for the verification-mode label. Normal and `-O` outputs are byte-identical; fresh discovery reproduces the certificate file byte-for-byte; the preceding simplified upper-97 result is reproduced unchanged. Fifteen negative controls pass, including the false reflected-margin shortcut, the invalid P2 ordinary shortcut, and an attempt to certify size 96. Actual commands, run times, input hashes, regeneration logs, and complete results are in `results/2026-10-16/upper96/`.

Proof of Theorem 1.1. The audited foundation excludes all cardinalities above 97. A hypothetical size-97 set contains zero and satisfies the lattice and scalar reduction. After the required contributor normalization, a high all-active state is either excluded by its valid oriented margin or assigned to an H0 group. All high one-empty states belong to H1. Low states belong to the six listed low families. Endpoint transport gives a necessary root model.

A root is either directly contradictory or covered by an exhaustive loss refinement. Any remaining integer instance follows a satisfied child of each proved disjunction; the transported endpoint arguments ensure that such a child exists. Every resulting leaf has a strict nonnegative integer-row contradiction. No size-97 feasible set exists, so $N \leq 96$. The existing 71-element construction supplies the lower bound. □

The next endpoint is not certified. For size 96, the balanced scalar values would be $d = 88$, $\Omega = 432$, $G = 164$, so the zero-hole cover cap is 268. This reaches $3d = 264$, permitting three empty units. With one empty unit the residual cap 180 also reaches $2(d-4) = 168$, permitting two large-loss centers. The present root universe and unique-large-loss rule therefore do not directly continue to $N \leq 95$. These are necessary-budget obstacles, not feasible constructions.

Reproduction. From the repository root, with Python 3.10 or later:

```
python -I -S -B verification/upper96.py
python -I -S -B verification/upper96.py --exhaustive
```

The optional generator requires SciPy:

```
python experiments/generate_upper96_certificates.py
```

The Windows entry `verify_upper96_windows.bat` runs the complete six-stage audited continuation from 115, using the simplified upper-97 step before this final exclusion. Reports and earlier certificates are retained unchanged for comparison.

References

[1] Sum-of-Subset research project, *A unified loss and hole proof from 98 to 97*, 16 October 2026. Repository report. Includes references to the audited preceding structural lemmas.

[2] Sum-of-Subset research project, *A shorter proof method for subset sums*, 16 October 2026. Repository report.

[3] Sum-of-Subset research project, *Complete upper-86 report*, 5 October 2026, sections on the size-97 exclusion and its disjunctions. Archived report. The present finite certificate data were generated independently.